

Quiz navigation

1 2 3 4 5

Finish review

Course Home

Course Info ^

Syllabus

Participants list

Grade/Attendance ^

Offline-Attendance

Grades

Students Notifications v

Others v

Activities/Resources +



Discrete Mathematics

7Week [23 March - 29 March]

Quiz7

Started on	2024-04-02 18:32
State	Finished
Completed on	2024-04-03 23:06
Time taken	1 day 4 hours
Grade	10.00 out of 10.00 (100%)

Question 1

Complete

Mark 3.00 out of 3.00

Find an inverse of 34 modulo 89 .

Write your solution in details and attach it below.

 IMG_20240403_225729.jpg

Comment:

Question 2

Correct

Mark 1.00 out of 1.00

If you know that -34 is the inverse of 34 modulo 89 , then solve the congruence $34x \equiv 21 \pmod{89}$.

Choose the correct answer(s) from below.

Select one or more:

- ☒ $87 \pmod{89}$ ✓
- ☒ $-2 \pmod{89}$ ✓
- ☒ $55 \pmod{89}$ ✗
- ☒ $-13 \pmod{89}$ ✗
- ☒ $-34 \pmod{89}$ ✗

Your answer is correct.

The correct answers are: $87 \pmod{89}$, $-2 \pmod{89}$

Question 3

Complete

Mark 2.00 out of 2.00

Use Fermat's little theorem to find $23^{1002} \pmod{41}$.

Attach your solution below.

 IMG_20240403_225738.jpg

Comment:

Question 4

Complete

Mark 3.00 out of 3.00

The encryption function is $e(x) = (7x + 3) \bmod 26$.

Decrypt the encrypted message "HQAD". The alphabet below may be helpful.

A	B	C	D	E	F	G	H	I	J	K	L	M
↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕
0	1	2	3	4	5	6	7	8	9	10	11	12

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕
13	14	15	16	17	18	19	20	21	22	23	24	25

Attach your solution below.

 IMG_20240403_225738.jpg

Comment:

Question 5

Correct

Mark 1.00 out of 1.00

Is the RSA system still secure, if we publish m and $\phi(m)$?

Select one:

- ☒ no ✓
- ☐ yes

Your answer is correct.

The correct answer is: no